

KEMPSEY, AustraliaWMO: **947850**Lat: **31.0711S**Lon: **152.7717E**Elev: **46**StdP: **14.67**Time Zone: **10.00 (AUS)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 34.9	(c) 37.4	(d) 27.6	(e) 21.6	(f) 56.2	(g) 30.3	(h) 24.5	(i) 52.7	(j) 21.1	(k) 62.3	(l) 18.7	(m) 62.4	(n) 4.0	(o) 250	(p) 0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 18.3	(c) 90.6	(d) 72.3	(e) 87.4	(f) 72.1	(g) 84.9	(h) 71.2	(i) 75.6	(j) 86.3	(k) 74.2	(l) 83.9	(m) 72.9	(n) 81.6	(o) 11.3	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	120.1	81.4	71.1	114.8	79.0	70.0	110.7	77.5	39.2	86.5	37.8	83.9	36.6	81.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.7	(b) 16.5	(c) 14.7	(e) 29.4	(f) 100.8	(g) 2.3	(h) 5.2	(i) 27.7	(j) 104.6	(k) 26.4	(l) 107.7	(m) 25.1	(n) 110.6	(o) 23.4	(p) 114.4		
			DB													
			WB													
				28.8			1.7									

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	73.9	73.3	71.0	65.8	59.6	56.2	54.2	55.6	60.4	64.4	68.4	71.2		
	DBStd	7.90	3.69	3.70	3.26	3.49	3.80	3.89	3.75	4.01	4.23	4.47	4.41	4.06		
	HDD50	13	0	0	0	0	0	3	7	3	0	0	0	0		
	HDD65	1330	0	0	1	30	171	265	336	293	150	66	15	3		
	CDD50	5282	740	653	651	475	297	189	136	177	311	445	551	657		
	CDD65	1123	275	233	187	55	3	0	0	2	11	46	116	195		
	CDH74	8184	2065	1584	1027	328	63	5	14	48	245	493	905	1407		
	CDH80	2249	672	486	219	40	2	0	0	9	64	116	246	395		
(14) Wind	WSAvg	6.1	6.4	6.3	5.7	5.6	5.2	5.8	5.9	6.4	6.5	6.4	6.6	6.8		
(15) Precipitation	PrecAvg	49.9	5.9	5.8	7.0	5.1	4.4	3.8	1.9	1.9	1.7	3.3	4.0	4.7		
	PrecMax	89.1	22.5	16.5	23.4	22.7	18.9	22.0	10.1	10.8	5.6	23.1	10.9	13.9		
	PrecMin	23.7	0.7	1.3	0.6	0.3	0.1	0.0	0.0	0.0	0.0	0.3	0.9	0.7		
	PrecStd	15.3	4.5	3.9	4.7	5.4	4.9	4.3	2.6	2.6	1.2	4.0	2.6	2.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.3	94.9	89.1	84.2	78.7	74.0	75.8	79.1	87.8	89.4	91.9	92.5		
		MCWB	73.7	76.3	73.3	67.8	62.3	59.1	57.8	57.6	62.2	66.3	68.8	71.3		
	2%	DB	89.7	88.7	85.1	80.5	75.7	70.1	71.1	74.0	80.4	82.8	86.0	87.7		
		MCWB	73.2	74.3	72.5	67.0	62.0	58.4	56.5	56.6	61.7	66.6	70.1	72.1		
	5%	DB	86.5	85.3	82.1	77.5	72.7	68.0	68.0	70.9	76.1	79.1	82.4	84.4		
		MCWB	72.9	72.7	70.8	65.6	60.6	57.2	55.3	56.2	60.5	65.5	69.0	70.7		
	10%	DB	83.6	82.5	79.7	75.3	70.1	65.7	65.2	68.0	72.7	76.0	79.1	81.5		
		MCWB	71.8	70.9	69.3	64.9	59.0	56.5	54.2	55.0	59.3	63.8	67.4	69.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.5	78.0	75.4	70.6	66.5	63.0	62.3	63.0	67.0	71.2	74.6	76.1		
		MCDB	89.0	90.2	85.5	79.4	74.4	68.3	68.4	73.1	77.9	82.1	86.0	87.5		
	2%	WB	75.5	75.7	73.5	68.8	64.1	61.4	59.6	60.1	64.7	68.7	72.2	73.8		
		MCDB	86.3	86.0	82.4	76.8	71.5	66.5	65.9	68.1	74.8	78.5	82.7	83.9		
	5%	WB	74.0	73.9	72.0	67.4	62.3	59.8	57.6	58.5	62.8	66.8	70.2	72.2		
		MCDB	83.4	82.9	79.8	74.8	69.6	65.0	64.5	66.8	72.4	76.1	79.2	81.4		
	10%	WB	72.8	72.3	70.7	66.0	60.8	58.3	55.9	56.7	61.0	65.2	68.6	70.8		
		MCDB	81.0	79.9	77.7	73.0	67.4	63.3	62.7	64.7	69.8	73.5	76.4	79.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	17.7	17.4	19.6	23.2	20.8	24.6	26.9	25.6	22.4	20.1	19.3		
		MCDBR	24.6	23.5	21.5	23.1	26.1	23.6	29.4	31.5	32.2	28.8	26.5	25.1		
	5% WB	MCWBR	11.2	10.5	10.5	11.3	13.5	13.3	15.8	16.3	15.8	14.8	13.1	11.5		
		MCWBR	20.2	20.9	19.8	20.1	21.6	17.3	21.8	25.8	26.9	24.5	22.1	21.3		
(40) Clear-Sky Solar Irradiance	taub		0.396	0.377	0.358	0.341	0.315	0.306	0.298	0.309	0.352	0.372	0.386	0.394		
		taud	2.470	2.547	2.590	2.604	2.629	2.647	2.654	2.608	2.476	2.457	2.470	2.464		
	Ebn at Noon		300	299	295	284	277	272	281	291	292	298	301	301		
		Edh at Noon	37	34	30	27	23	22	23	26	33	36	37	38		
(44) All-Sky Solar Radiation	RadAvg		2058	1831	1550	1298	1050	867	991	1285	1649	1841	1951	2052		
	RadStd		152	150	126	86	75	60	79	89	109	161	192	178		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-99	+263	N/A

Nomenclature: See separate page